

Multiomi

(Ex 22)

9) $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$

$$A \cap (B \cup C) \stackrel{\text{def}}{=} A \cap ((B \cap C) \cup (C \setminus B)) \stackrel{X \cap Y = X \cap C_Y}{=} A \cap ((B \cap C) \cup (C \setminus B))$$

$$= A \cap ((B \cap C) \cup (C \cap B)) =$$

$$\xrightarrow{\text{distr}} (A \cap B \cap C) \cup (A \cap (B \cap C)) = \text{FND}$$

$$(A \cap B) \Delta (A \cap C) = \cancel{((A \cap B) \cup (A \cap C))} \setminus$$

$$= ((A \cap B) \cap (A \cap C)) \cup ((A \cap C) \cap (A \cap B))$$

De Morgan $\overline{(A \cap B)} = (\overline{A} \cup \overline{B})$

$$(A \cap C) \cap (A \cup B) =$$

$$\stackrel{\text{Distrib.}}{=} (A \cap B \cap (C \cup D)) \cup (A \cap C \cap (C \cup D))$$

$$= ((A \cap B \cap C_A) \cup (A \cap B \cap C_C)) \cup$$

$$(A \cap C \cap C_A) \cup (A \cap C \cap C_B) =$$

$$C_{A \cap A} = (\underbrace{(A \cap B)}_{\text{red}}) \cup (A \cap B \cap C) \cup$$

$$\sum_{\emptyset} ((\emptyset \cap C) \cup (A \cap C \cap C_B)) =$$

$$= (A \cap B \cap C_C) \cup (A \cap C \cap C_B)$$

(Ex 3.4) Se se dă o pt & mulțimi A, B, C dacă
 $A \cap C = B \cap C$ (1) și $A \cup C = B \cup C$ (2) atunci $A = B$

Fie $x \in A$ arbitrar

$$x \in A \Rightarrow x \in A \cup C \xrightarrow{(2)} x \in B \cup C \xrightarrow[\text{"U"}]{\text{def}} x \in B \vee x \in C$$

(CI) $x \in B \xrightarrow{x \in A} A = B \quad (\forall x \in A, x \in B \Leftrightarrow A \subseteq B)$

(CI) $\left. \begin{array}{l} x \in C \\ x \in A \end{array} \right\} \Rightarrow x \in A \cap C \quad (1) \Rightarrow x \in B \cap C \Leftrightarrow$

$$x \in B \wedge x \in C \Rightarrow \Rightarrow A \subseteq B \Bigg\} \Rightarrow$$

De simetrie $B \subseteq A$

$$\rightarrow A = B$$

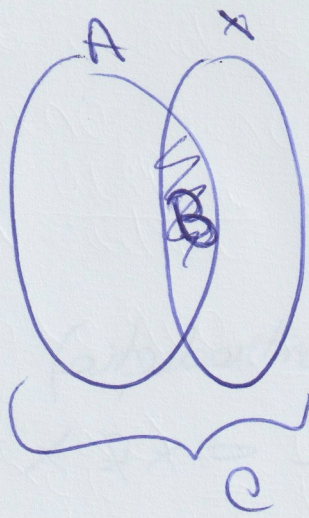
(MII) $A \stackrel{\text{absorbție}}{=} A \cap (A \cup C) \xrightarrow{(2)} A \cap (B \cup C) =$
 $= (A \cap B) \cup (A \cap C) \stackrel{(1)}{=} (A \cap B) \cup (B \cap C) = B$

! (factor comun) !

$$= B \cap (A \cup C) = B \cap (B \cup C) \stackrel{\text{absorbție}}{=} B$$

(Ex 3.5) Fie A, B, C mulțimi date. Se se dă mult x
 care satisface:

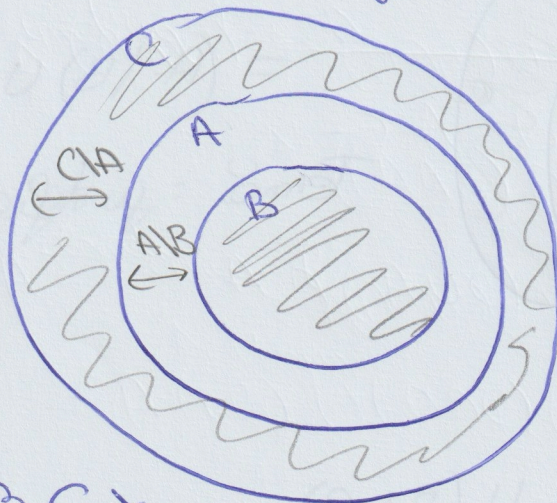
o) $A \cap x = B, A \cup x = C$



$$X = (C \setminus A) \cup B$$

$$(1) A \cap X = B \Rightarrow \begin{cases} B \subseteq A \\ B \subseteq X \end{cases} \Rightarrow B \subseteq A \subseteq C$$

$$(2) A \cup X = C \Rightarrow \begin{cases} A \subseteq C \\ X \subseteq C \end{cases}$$



$$1. B \subseteq X$$

$$2. \text{ For } x \in A \setminus B \Rightarrow x \in A \text{ and } x \notin B$$

$$\begin{aligned} \text{If } \begin{cases} x \in A \\ x \in X \end{cases} &\Rightarrow x \in A \cap X \stackrel{(1)}{\Rightarrow} x \in B \quad \text{FALSE} \\ &\Rightarrow x \notin X \end{aligned}$$

And/or $x \in A \setminus B$ disjoint

$$3. \text{ For } x \in C \setminus A \Rightarrow x \in C \text{ and } x \notin A$$

$$\begin{aligned} x \in C &\stackrel{(2)}{\Rightarrow} x \in A \cup X \Rightarrow \begin{cases} x \in A \text{ or } x \in X \\ x \notin A \end{cases} \\ &\Rightarrow x \in X \end{aligned}$$

Assoder $C/A \subseteq X$

4. C_c

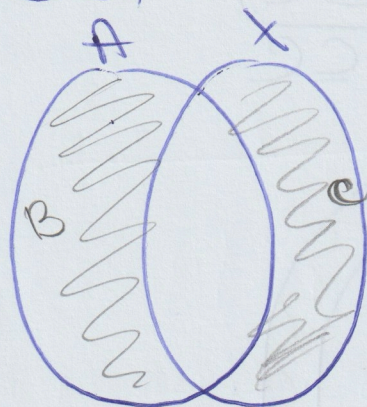
Für $x \in C_c \Rightarrow x \notin C$ (Contradiction)

oder $X \subseteq C \Rightarrow x \notin X \Rightarrow C_c$ ist

somit disjunkt

Assoder $X = (C/A) \cup B$

g) $\begin{cases} A \setminus X = B & (1) \\ X \setminus A = C & (2) \end{cases}$



$$X = (A \cup C) \setminus B = (A \setminus B) \cup C$$

Lemma: justifizieren

(Ex 3.6)

g) $(A \cup B) \times C = (A \times C) \cup (B \times C)$

Für (a, b) beliebig

$$(a, b) \in (A \cup B) \times C \stackrel{\text{def}}{\iff} (a \in (A \cup B) \wedge b \in C) \stackrel{\text{"U"}}{\iff}$$

$$\iff (a \in A \vee a \in B) \wedge b \in C$$

$$\stackrel{\text{distrib}}{\iff} (a \in A \wedge b \in C) \vee (a \in B \wedge b \in C)$$

$$\stackrel{\text{"X"}}{\iff} ((a, b) \in A \times C) \vee ((a, b) \in B \times C)$$

$$\iff ((a, b) \in A \times C) \cup ((a, b) \in B \times C)$$

SESIONE

$$\Rightarrow (A \times C) \cup (B \times C) \quad (\text{ANDEVARANT})$$

$$e) (A \setminus B) \times C = (A \times C) \setminus (B \times C)$$